Let's begin by labeling the possible spin states along the three coordinate axes. If $\mathcal{A}$ is oriented along the *z* axis, the two possible states that can be prepared correspond to $\sigma_z = \pm 1$. Let's call them *up* and *down* and denote them by ket-vectors $|u\rangle$ and $|d\rangle$. Thus, when the apparatus is oriented along the *z* axis and registers +1, the state $|u\rangle$ has been prepared.

On the other hand, if the apparatus is oriented along the *x* axis and registers −1, the state $|l\rangle$ has been prepared. We'll call it *left*. If $\mathcal{A}$ is along the *y* axis, it can prepare the states $|i\rangle$ and $|o\rangle$ (*in* and *out*). You get the idea.

The idea that there are no hidden variables has a very simple mathematical representation: the space of states for a single spin has only two dimensions. This point deserves emphasis:

All possible spin states can be represented in a two-dimensional vector space.

We could, somewhat arbitrarily,[1] choose $|u\rangle$ and $|d\rangle$ as the two basis vectors and write *any* state as a linear superposition of these two. We'll adopt that choice for now. Let's use the symbol $|A\rangle$ for a generic state. We can write this as an equation,

$$|A\rangle = \alpha_u|u\rangle + \alpha_d|d\rangle,$$

where α_u and α_d are the components of $|A\rangle$ along the basis directions $|u\rangle$ and $|d\rangle$. Mathematically, we can identify the components of $|A\rangle$ as

$$\begin{aligned} \alpha_u &= \langle u|A\rangle \\ \alpha_d &= \langle d|A\rangle. \end{aligned} \qquad (2.1)$$

These equations are extremely abstract, and it is not at all obvious what their physical significance is. I am going to tell you right now what they mean: First of all, $|A\rangle$ can represent any state of the spin, prepared in any manner. The components α_u and α_d are complex numbers; by themselves, they have no experimental meaning, but their magnitudes do. In particular, $\alpha^*_u\alpha_u$ and $\alpha^*_d\alpha_d$ have the following meaning: